

Минимальная временная корневая файловая система

1. Накидываем прогу

```
#include <stdio.h>
#include <unistd.h>

int main(void)
{
    printf("HELL0000000000o!!!!\n");
    sleep(20);
}
```

2. Компилируем, используя `–static` для статического применения части библиотек в бинарнике.

```
root@kivvi-VirtualBox:/home/kivvi/tests/armtest# arm-linux-gnueabihf-gcc -static init.c -o init
```

3. Используем команду для создания простой временной файловой системы

```
armtest# echo init | cpio -o -H newc | gzip > initramfs.cpio.gz
```

4. Запускаем виртуализацию:

```
root@kivvi-VirtualBox:/home/kivvi/tests/armtest# QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0" -nographic
```

5. Ядро запустилось, 20 секунд пожило и полегло:

```
[ 2.054840] Freeing unused kernel image (initmem) memory: 2040k
[ 2.700237] Run /init as init process
HELL0000000000o!!!!
[ 12.863544] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 12.863629] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
[ 12.863642] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
[ 12.863654] amba 100e5000.watchdog: deferred probe pending: (reason unknown)
[ 22.848402] Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000
[ 22.849899] CPU: 0 UID: 0 PID: 1 Comm: init Not tainted 7.1.0-rc5-00149-g8fde5d1d47f6 #1 VOLUNTARY
[ 22.850323] Hardware name: ARM-Versatile Express
[ 22.850661] Call trace:
[ 22.851255] unwind_backtrace from show_stack+0x10/0x14
[ 22.852255] show_stack from dump_stack_lvl+0x54/0x68
[ 22.852482] dump_stack_lvl from vpanic+0x234/0x454
[ 22.852722] vpanic from panic+0x2c/0x30
[ 22.852972] panic from do_exit+0xa44/0xa48
[ 22.853248] do_exit from do_group_exit+0x34/0x84
[ 22.853443] do_group_exit from pid_child_should_wake+0x0/0x60
[ 22.854529] ---[ end Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000 ]---
```

Busybox

1. Делаем дефолтный конфиг

```
/busybox-1.38.0# ARCH=arm make defconfig
```

2. Запускаем меню конфига

```
/busybox-1.38.0# ARCH=arm make menuconfig
```

Где переходим в Settings и выставляем Build static binary – yes и Cross compiler prefix – arm-linux-gnueabihf-

```
--- Build Options
[*] Build static binary (no shared libs)
[ ] Force NOMMU build

--- Cross compiler prefix
(arm-linux-gnueabihf-) Cross compiler prefix
( ) Path to crossroot
```

3. Запускаем сборку

```
/busybox-1.38.0# make
```

В результате сборки возникают ошибки, связанные с networking/tc. Отключим его и попробуем снова.

```
networking/tc.c:300:32: error: 'TCF_CBQ_LSS_ISOLATED' undeclared (first use in this function)
300 |         if (lss->flags&TCF_CBQ_LSS_ISOLATED) {
    |                                ^~~~~~
networking/tc.c:308:24: error: invalid use of undefined type 'struct tc_cbq_wrropt'
308 |         if (wrr->priority != TC_CBQ_MAXPRIO)
    |         ^~
networking/tc.c:308:38: error: 'TC_CBQ_MAXPRIO' undeclared (first use in this function)
308 |         if (wrr->priority != TC_CBQ_MAXPRIO)
    |                                ^~~~~~
networking/tc.c:309:46: error: invalid use of undefined type 'struct tc_cbq_wrropt'
309 |         printf("prio %u", wrr->priority);
    |                                ^~
networking/tc.c:313:43: error: invalid use of undefined type 'struct tc_cbq_wrropt'
313 |         printf("/%u ", wrr->cpriority);
    |                                ^~
networking/tc.c:314:32: error: invalid use of undefined type 'struct tc_cbq_wrropt'
314 |         if (wrr->weight != 1) {
    |         ^~
networking/tc.c:315:65: error: invalid use of undefined type 'struct tc_cbq_wrropt'
315 |         print_rate(buf, sizeof(buf), wrr->weight);
    |                                ^~
networking/tc.c:318:32: error: invalid use of undefined type 'struct tc_cbq_wrropt'
318 |         if (wrr->allot)
    |         ^~
networking/tc.c:319:57: error: invalid use of undefined type 'struct tc_cbq_wrropt'
319 |         printf("allot %ub ", wrr->allot);
```

4. В этот раз сборка проходит успешно:

```
root@kivvi-VirtualBox:/home/kivvi/prog/kernel/fs/busybox-1.38.0# file busybox
busybox: ELF 32-bit LSB executable, ARM, EABI5 version 1 (GNU/Linux), statically linked, BuildID[sha1]=8894231ebdbed33ff9d79f18ede7dad1f51dfa8a, for GNU/Linux 3.2.0, stripped
```

5. Установка

```
stripped
busybox-1.38.0# make install
```

6. Создание архива с файловой системой

```
root@kivvi-VirtualBox:/home/kivvi/prog/kernel/fs/busybox-1.38.0/_install# find . | cpio -o -H newc | gzip > initramfs.cpio.gz
```

7. Запускаем виртуализацию

```
root@kivvi-VirtualBox:/home/kivvi/tests/armtest# QEMU AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0 rdinit=/bin/ash" -nographic
```

8. Всё работает:

```
[ 3.164231] clk: Disabling unused clocks
[ 3.164806] PM: genpd: Disabling unused power domains
[ 3.222969] Freeing unused kernel image (initmem) memory: 2048K
[ 3.228457] Run /bin/ash as init process
/bin/ash: can't access tty; job control turned off
~ # ls
bin          initramfs.cpio.gz  root         usr
dev          linuxrc            sbin
~ # [ 13.306599] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 13.306682] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
[ 13.306696] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
[ 13.306708] amba 100e5000.watchdog: deferred probe pending: (reason unknown)
ls
bin          initramfs.cpio.gz  root         usr
dev          linuxrc            sbin
~ # whoami
whoami: unknown uid 0
~ # ls -la
total 1028
drwxr-xr-x  7 0      0          200 Jun  1 14:16 .
drwxr-xr-x  7 0      0          200 Jun  1 14:16 ..
-rw-r--r--  1 0      0           17 Jun  1 14:19 .ash_history
drwxr-xr-x  2 0      0         1920 Jun  1 14:00 bin
drwxr-xr-x  2 0      0           60 May 29 12:59 dev
-rw-r--r--  1 0      0       1048576 Jun  1 14:11 initramfs.cpio.gz
lrwxrwxrwx  1 0      0           11 Jun  1 14:00 linuxrc -> bin/busybox
drwx-----  2 0      0           40 May 29 12:59 root
drwxr-xr-x  2 0      0         1460 Jun  1 14:00 sbin
drwxr-xr-x  4 0      0           80 Jun  1 14:00 usr
~ # history
0 ls
1 whoami
2 ls -la
3 history
~ #
```